HVDC Transmission Introduction

ABSTRACT:

This report briefly discusses HVDC transmission systems, the components and converter topologies, advantages of one over the other, the system topologies and a few papers regarding the same

HVDC transmission- High voltage DC transmission is a method of power transmission which uses DC power instead of AC power to transmit electrical power over long distances
DC has a lot of advantages over AC when it comes to transmitting [1]:

- All the power transmitted is used, unlike in AC where the out of the power transmitted, only the reactive part is actually used
- Clearance between phases and grounds can be lower
- Lower number of conductors required for transmission power
- Lower mechanical load on DC transmission powers and thus smaller tower
- Cost of transmission line per unit distance is smaller
- No losses due to radiation, inductive, capacitive skin effect or corona effects greatly increasing efficiency of an HVDC transmission system
- For the same power delivered, the conductor size for DC transmission can be smaller due to there being no reactive power component

HVDC systems have 2 major different configurations, the LCC and the VSC modes of operation with the latter being more recent and efficient
Before talking about the two modes, the basic structure of a transmission system is like follows in Figure 1: HVDC transmission
The AC bus is connected to a step up transformer which converts it to high voltage (varies from 33kV to 200kV and beyond) which is then passed to a rectifier consisting of either thyristors or IGBTs depending on the configuration being used
The transmission then occurs over a long distance (1000km etc in india connecting northeast, bhutan and agra) after which it is converted back to AC using either multilevel converters or thyristors and used to create 3 phase high voltage AC which can be stepped down and supplied to houses
The rectifying and inverting components in Figure 1: LCC transmission system is called a thyristor which is basically a diode which can be turned on by an external signal (often connected to an ac supply to time the opening and closing of the diode), these components (thyristor and IGBTs) are often called valves
In the paper (Bahrman and Johnson 2007)[5] the topic of LCC and VSC and the different components they use is discussed in detail enough providing the reader a good enough description to understand the topic at hand
The paper starts off by discussing LCC and VSCs referring to the differences in control and commutation (switching off of the device) and talks a bit about the history of these two modes of HVDC transmission application
Then it moves on to the technical discussion between these two modes

The LCC (line commutated current source converter) uses a current source (battery parallel to inductor) and multiple thyristor modules in an H bridge configuration to produce a pulse modulated wave for which if the voltage was averaged, the values would be of the form of a sine wave. This as said by the paper is often done by connecting the signal line of the thyristor to a similar AC signal as desired on the output to ensure synchronicity
Figure 2: LCC HVDC
The AC to DC side uses thyristors as normal "diodes" to rectify the AC power and the inverter is synchronised with an existing AC signal

The VSC (Voltage source converter) uses a similar configuration but it instead uses IGBT cells for rectification and inversion of AC and DC power
The VSC uses IGBTs (insulated gate bipolar transistors) with a very high collector emitter breakdown voltage allowing for even higher voltage transmissions to transmit power
In a similar structure to the LCC configuration, basically an IGBT rectifier bridge is used to convert stepped up high voltage AC (even 800kV in india) to high voltage DC and transmit over a long distance where it is converted back to AC by the use of pulse width modulation and IGBT with a control system to give out a highly modulated sine wave which is then smoothed by using passive or active filters
The control is usually done by comparing existing sine waves to a sawtooth wave and switching the particular IGBT on and off to create a PWM signal with multiple levels depending on the configuration of the converter used to simulate a sine wave
This is then simply filtered to provide a sinusoidal signal
VSC has the advantage of faster switching times and higher voltage tolerance which can even be further increased by increasing the number of IGBT cells in series and increased current thresholds by increasing the amount of IGBT cells in parallel
The paper then moves on to the different types of transmission systems that are used, including monopole, bipole and multiterminal
The paper also discusses the different types of configurations of the above given types as shown in Figure 3: Types of Transmission Systems
Here Bipolar implies that the generated HVDC is split into two, a positive and negative voltage which is transmitted. This also helps in maintaining a return path and easier implementation of multiterminal systems while monopolar is the cheap and simpler implementation using a single pole to transmit power. The ground return and metallic return are the two types for monopole and bipole, ground return being that ground or the earth is used as the return path for the current, this however is not viable in all scenarios, especially where the earth has high resistivity, so the second type, the metallic return is used to avoid this by creating a path of return with a very low resistivity value, here the ground is used only as a reference potential
The paper then discusses the physical layout of an HVDC transmission system talking about the placement of capacitor banks, valve arrestors, cooling systems and how the design of the station is done taking care of the flow of electricity between components and safety of personnel working in the station Figure 4: Generic VSC HVDC Station
And in the end, the paper discusses the objectives of such a system:

1. to control basic system quantities such as dc line current, dc voltage, and transmitted power accurately and with sufficient speed of response
2. to maintain adequate commutation margin (switching off time) in inverter operation so that the valves can recover their forward blocking capability after conduction before their voltage polarity reverses

3. to control higher-level quantities such as frequency in isolated mode or provide power oscillation damping to help stabilize the ac network
4. to compensate for loss of a pole, a generator, or an ac transmission circuit by rapid readjustment of power
5. to ensure stable operation with reliable commutation in the presence of system disturbances
6. to minimize system losses and converter reactive power consumption
7. to ensure proper operation with fast and stable recoveries during ac system faults and disturbances

The second paper (Long and Nilsson 2007) discusses the various advancements in the field from 1930s starting with mercury filled arc valves and ending with MMC which are converters using multitudes of IGBT cells to switch which help achieve waveforms very close to a pure sine wave and the vice versa with high current and voltage capacities

The paper starts the discussion with the early research efforts around the world in the 1930s when the need for DC transmission lines was getting significant but the technology was not sufficiently advanced to allow for such a system

It then moves on to the 1950s when ASEA (sweden) made the first ever 20MW HVDC system because the region of Gotland had no local power generation, this laid the steps for future innovation

By the 1970s, thyristors and computers were taking control of HVDC stations which previously used operators in harsh environments and mercury arc valves

Since then, the systems have undergone multiple developments including the usage of IGBTs using cascaded H bridges to the MMC which uses multiple IGBT cells to achieve high voltages using low voltages

The paper ends by discussing the need of HVDC systems in china and India where power sparsity is a significant issue (and is being tackled now by the multiple HVDC power lines in india with some of them being considered very long and ultra high voltage (1400 km+ and 800kV +) and talks about how innovation is further required in this field to facilitate much more efficient power transmission

The next paper under consideration (Nami and Nademi 2018) (and still being read through properly) talks about the converter discussed in the previous papers, namely a version of the VSC which uses multiple cells of IGBTs to produce and rectify AC voltages

This paper starts off with the design, constraints and mathematical models of the MMC circuit showing how current and voltages vary in a generic MMC circuit and what the design constraints are. A lot of mathematical equations and relations are discussed before finally discussing the various topologies of the MMC circuit

The paper talks about the different control schemes and their effects on the outputs and ends with discussing emerging applications and alternatives to the current MMC and future trends

Websites:

1. https://www.electricaltechnology.org/2020/06/difference-between-hvac-hvdc.html
2. https://news.energysage.com/black-start-why-it-matters/
3. https://www.youtube.com/watch?v=ccKw9-FQOqw

Figures:

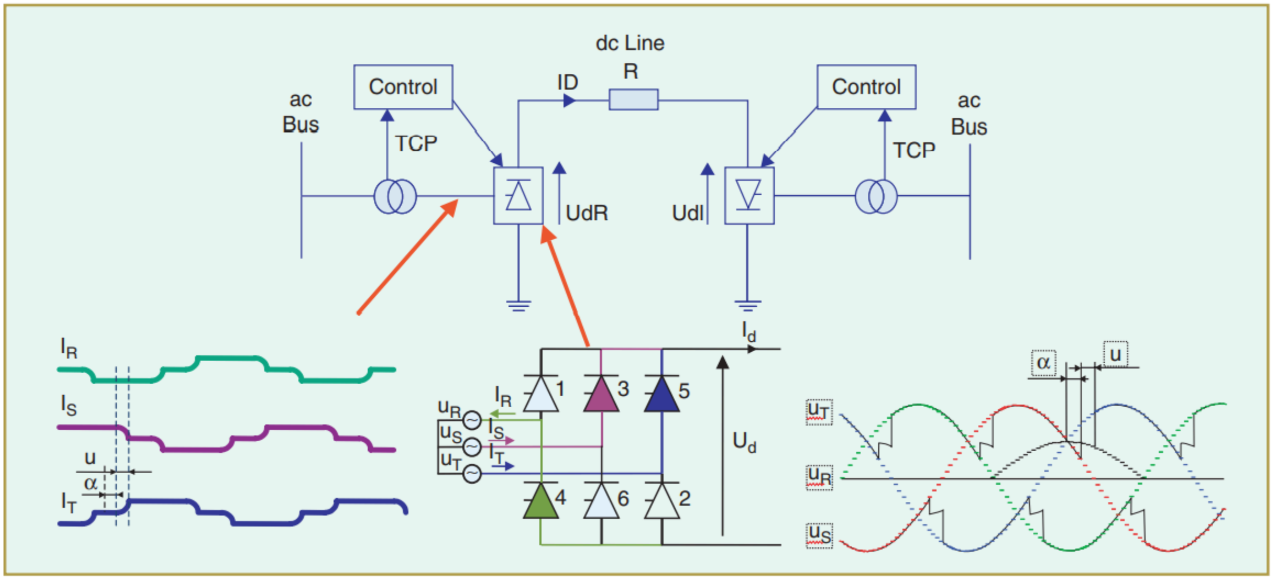

Figure 1: HVDC transmission

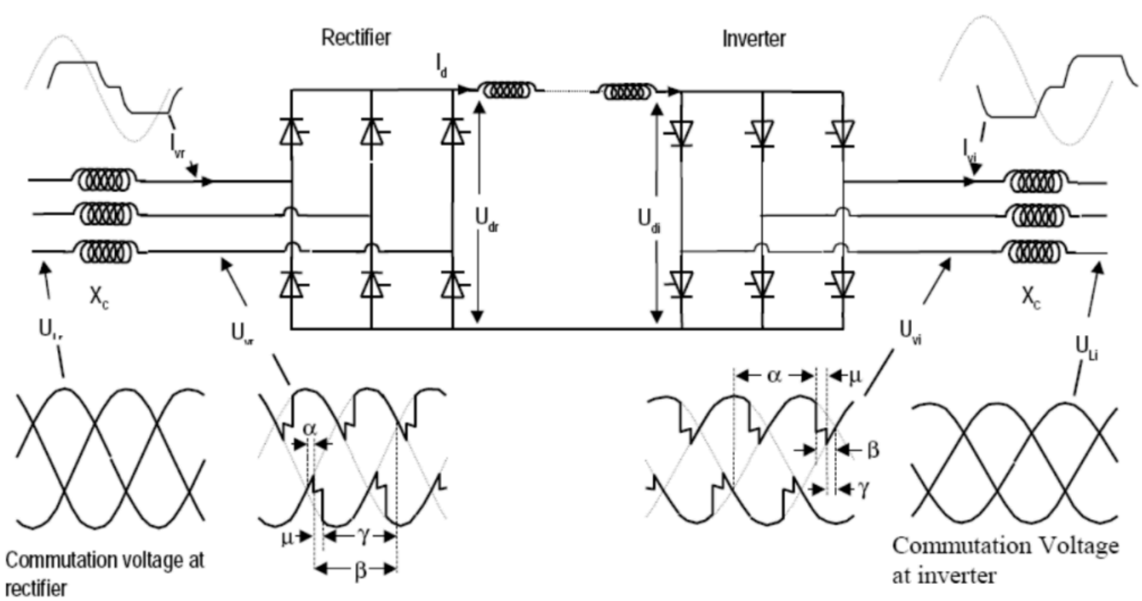

Figure 2: LCC HVDC

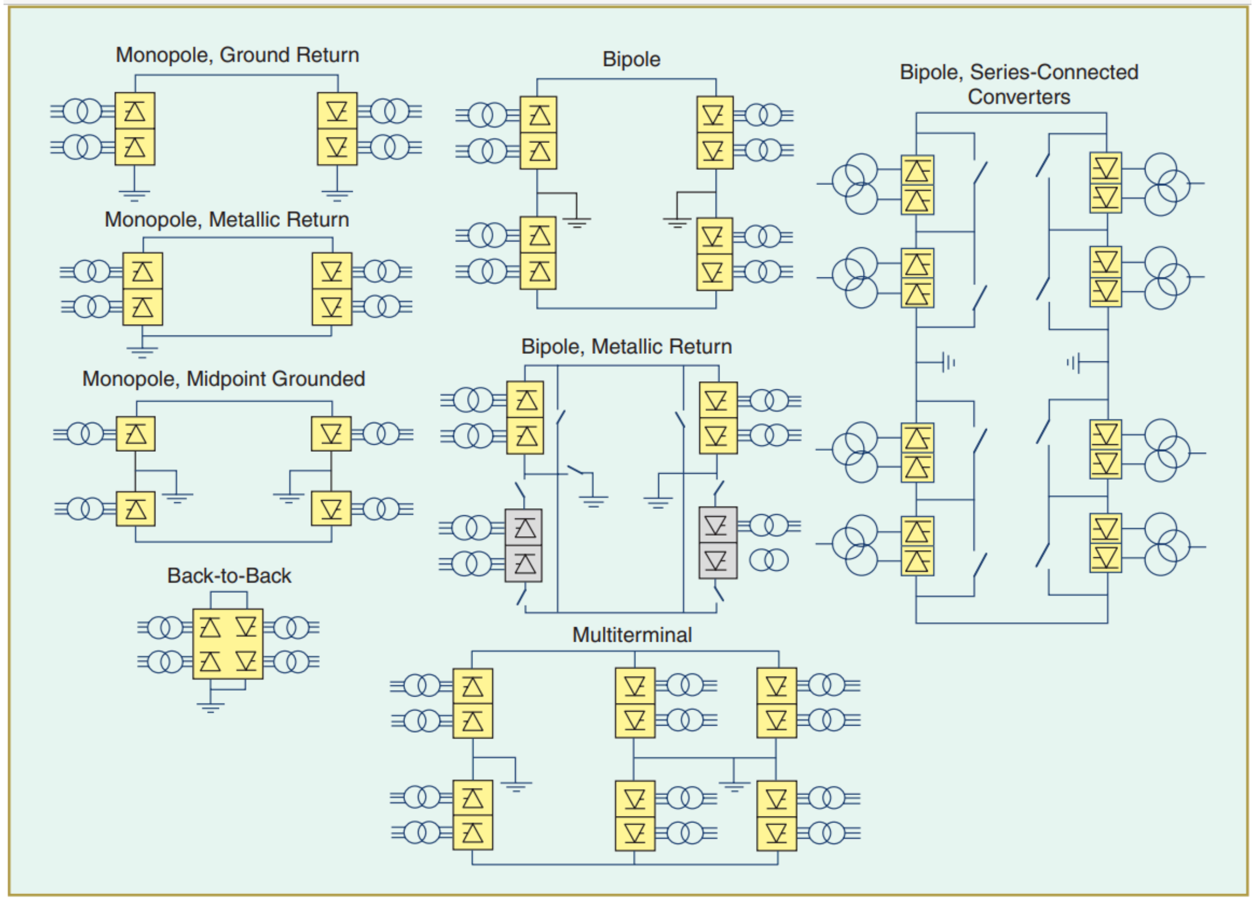

Figure 3: Types of Transmission Systems

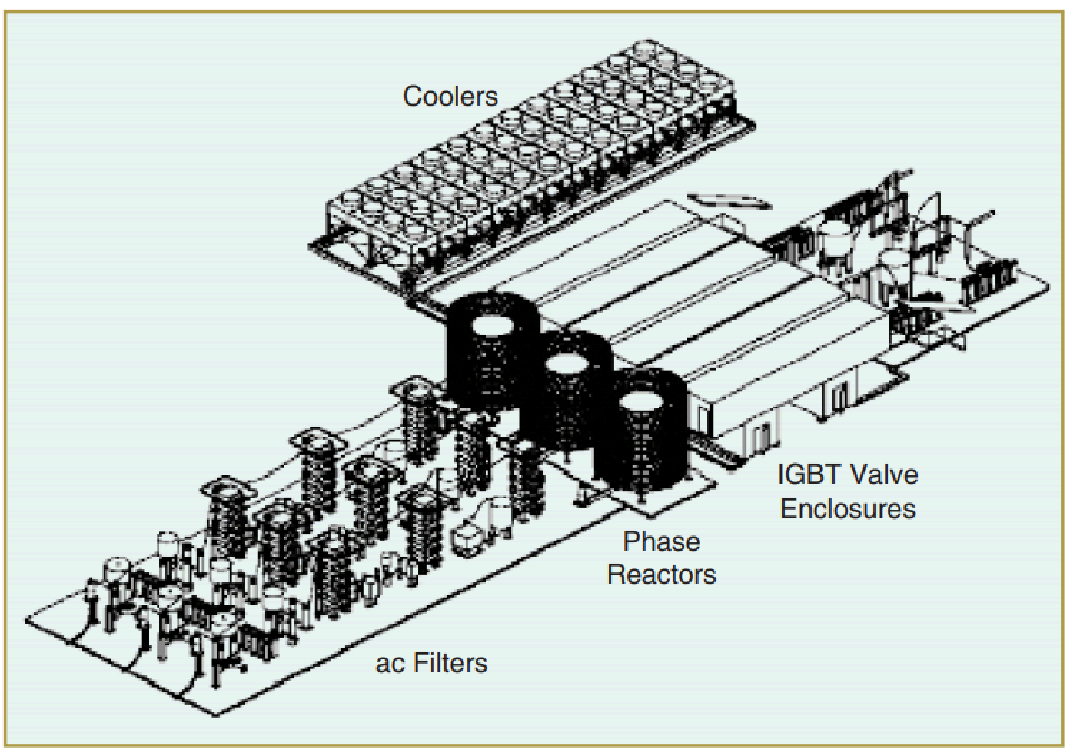

Figure 4: Generic VSC HVDC Station